TUTORIAL 2: KINEMATICS OF A PARTICLE

ME 1060: DYNAMICS

Question 1

The curvilinear motion of a particle is defined by $v_x = 50 - 16t$ and $y = 100 - 4t^2$, where v_x is in meters per second, y is in meters, and t is in seconds. It is also known that $x = 0$ when $t = 0$. Plot the path of the particle and determine its velocity and acceleration when the position $y = 0$ is reached.

Solution

The x-coordinate is obtained by integrating the expression for v_x, and the x-component of the acceleration is obtained by differentiating v_x. Thus,

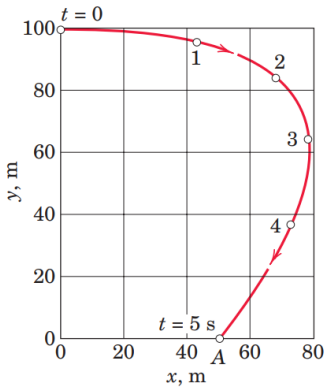
$$\left[\int dx = \int v_x dt\right] = \int_0^x dx = \int_0^t (50 - 16t)\, dt = 50t - 8t^2\ m$$

$$[a_x = \dot{v}_x] \qquad a_x = \frac{d(50 - 16t)}{dt} = -16 m/s^2$$

The y-components of velocity and acceleration are

$$[v_y = \dot{y}] \qquad v_y = \frac{d}{dt}(100 - 4t^2) = -8t\ \ m/s$$

$$[a_y = \dot{v}_y] \qquad a_y = \frac{d}{dt}(-8t) = -8\ m/s^2$$

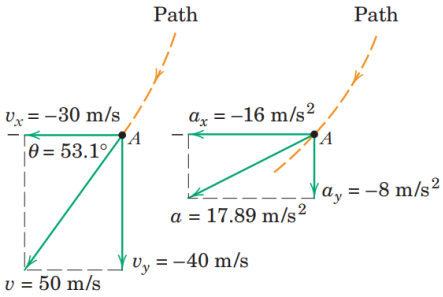


We now calculate corresponding values of x and y for various values of t and plot x against y to obtain the path as shown.

When $y = 0$, $0 = 100 - 4t^2$. Therefore $t = 5s$. For this value of the time, we have,

$$v_x = 50 - 16(5) = -30\ m/s$$

$$v_y = -8(5) = -40\ m/s$$



$$v = \sqrt{(-30)^2 + (-40)^2} = 50\ \ m/s$$

$$a = \sqrt{(-16)^2 + (-8)^2} = 17.89\ \ m/s^2$$

The velocity and acceleration components and their resultants are shown on the separate diagrams for point A, where $y = 0$. Thus, for this condition we may write

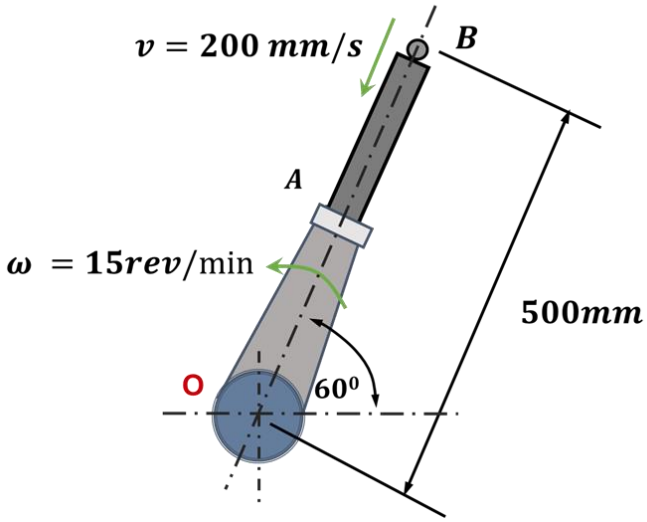
$$\boldsymbol{v} = -30\boldsymbol{i} - 40\boldsymbol{j}\ m/s$$

$$\boldsymbol{a} = -16\boldsymbol{i} - 8\boldsymbol{j}\ m/s^2$$

Question 2

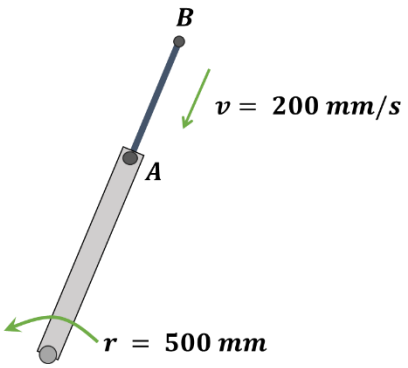
The motion of cutting tool *B* is controlled by the robot arm *OAB* as shown in Figure 2.

At the instant shown, the arm rotates anticlockwise at uniform angular speed $\omega = 15\ rev/min$ and the length of portion AB decreases at the uniform rate of $200mm/s$. If the length of *OB* at this instant is $500mm$, determine the velocity and acceleration of *B*.



Solution

Velocity of B;

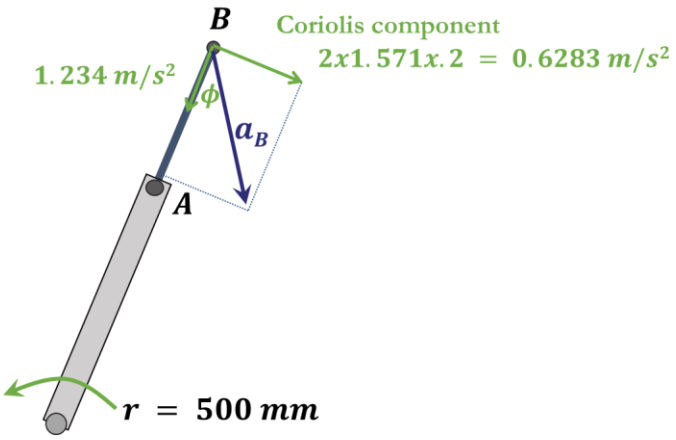


$$v_r = \dot{r} = \frac{0.2m}{s},$$

$$v_\theta = r\dot{\theta} = \frac{\left(0.5 \times \frac{2\pi}{60} \times 15\right) m}{s} = \frac{0.7854m}{sv_r} = \frac{0.2m}{sv_B} = \sqrt{{v_r}^2 + {v_\theta}^2} = \underline{\frac{0.8105m}{s}}$$

$$\varphi = \tan^{-1}\left(\frac{r\dot{\theta}}{\dot{r}}\right) = \underline{75.7^0}$$

Acceleration



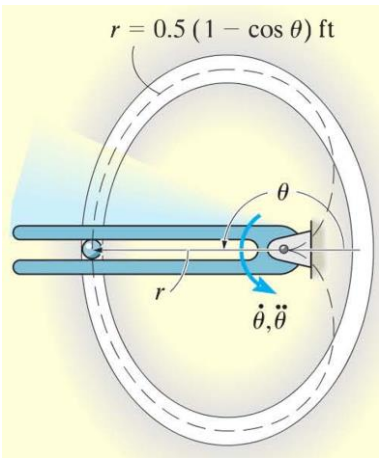
Acceleration of B

$a_B = 1.3844\, m/s2$

At 26.980 to AB

Question 3

Due to the rotation of the forked rod, the ball in figure travels around the slotted path, a portion of which is in the shape of a cardioid, $r = 0.5(1 - cos\,\theta)$ ft, where θ is in radians. If the ball's velocity is $v = 4\,ft/s$ and its acceleration is $a = 30\,ft/s^2$ at the instant. $\theta = 180°$, determine the angular velocity $\dot{\theta}$ and angular acceleration $\ddot{\theta}$ of the fork.



Solution

This path is most unusual, and mathematically it is best expressed using polar coordinates, as done here, rather than rectangular coordinates. Also, since $\dot{\theta}$ and $\ddot{\theta}$ must be determined, then r, θ coordinates are an obvious choice.

Velocity and Acceleration. The time derivatives of r and θ ca determined using the chain rule.

$$r = 0.5(1 - \cos\theta)$$
$$\dot{r} = 0.5(\sin\theta)\dot{\theta}$$
$$\ddot{r} = 0.5(\cos\theta)\dot{\theta}(\dot{\theta}) + 0.5(\sin\theta)\ddot{\theta}$$

Evaluating these results at $\theta = 180°$, we have

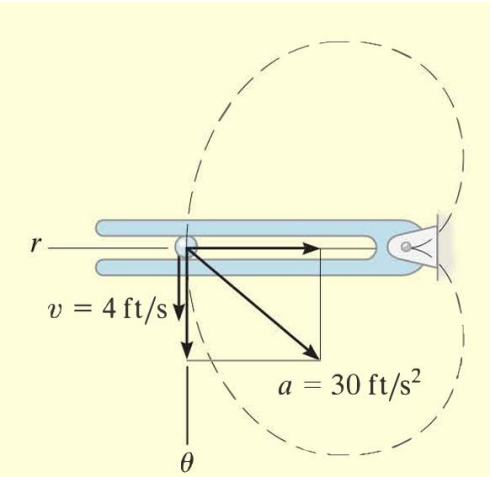
$$r = 1 \text{ ft} \qquad \dot{r} = 0 \qquad \ddot{r} = -0.5\dot{\theta}^2$$

Since $v = 4$ ft/s, using Eq. 12–26 to determine $\dot{\theta}$ yields

$$v = \sqrt{(\dot{r})^2 + (r\dot{\theta})^2}$$
$$4 = \sqrt{(0)^2 + (1\dot{\theta})^2}$$
$$\dot{\theta} = 4 \text{ rad/s}$$

In a similar manner, $\ddot{\theta}$ can be found using Eq. 12–30.

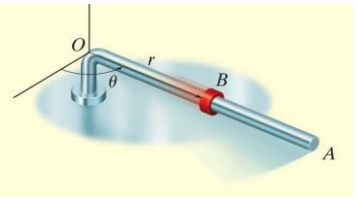
$$a = \sqrt{(\ddot{r} - r\dot{\theta}^2)^2 + (r\ddot{\theta} + 2\dot{r}\dot{\theta})^2}$$
$$30 = \sqrt{[-0.5(4)^2 - 1(4)^2]^2 + [1\ddot{\theta} + 2(0)(4)]^2}$$
$$(30)^2 = (-24)^2 + \ddot{\theta}^2$$
$$\ddot{\theta} = 18 \text{ rad/s}^2 \qquad \textit{Ans.}$$



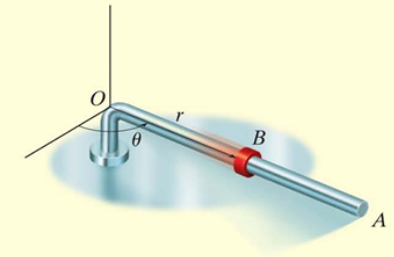
At this location, the θ and t (tangential) axes will coincide. The +n (normal) axis is directed to the right, opposite to $+r$.

Question 4

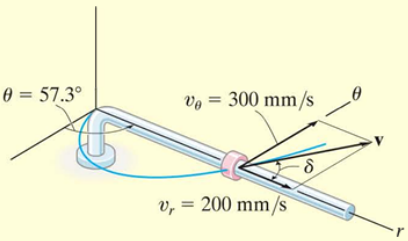
The rod OA in Figure rotates in the horizontal plane such that $\theta = t^3\ rad$. At the same time, the collar B is sliding outward along OA so that $r = 100\ t^2$ mm. If in both cases t is in seconds, determine the velocity and acceleration of the collar when $t\ =\ 1\ s$.



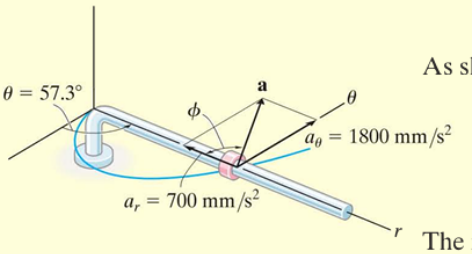
Solution



(a)



(b)



(c)

Fig. 12–33

SOLUTION

Coordinate System. Since time-parametric equations of the path are given, it is not necessary to relate r to θ.

Velocity and Acceleration. Determining the time derivatives and evaluating them when $t = 1$ s, we have

$$r = 100t^2\Big|_{t=1\,s} = 100\,\text{mm} \quad \theta = t^3\Big|_{t=1\,s} = 1\ \text{rad} = 57.3°$$

$$\dot{r} = 200t\Big|_{t=1\,s} = 200\ \text{mm/s} \quad \dot{\theta} = 3t^2\Big|_{t=1\,s} = 3\ \text{rad/s}$$

$$\ddot{r} = 200\Big|_{t=1\,s} = 200\ \text{mm/s}^2 \quad \ddot{\theta} = 6t\Big|_{t=1\,s} = 6\ \text{rad/s}^2.$$

As shown in Fig. 12–33*b*,

$$\mathbf{v} = \dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta$$
$$= 200\mathbf{u}_r + 100(3)\mathbf{u}_\theta = \{200\mathbf{u}_r + 300\mathbf{u}_\theta\}\ \text{mm/s}$$

The magnitude of $\mathbf{v}$ is

$$v = \sqrt{(200)^2 + (300)^2} = 361\ \text{mm/s} \qquad \textit{Ans.}$$

$$\delta = \tan^{-1}\left(\frac{300}{200}\right) = 56.3° \qquad \delta + 57.3° = 114° \qquad \textit{Ans.}$$

As shown in Fig. 12–33*c*,

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{u}_\theta$$
$$= [200 - 100(3)^2]\mathbf{u}_r + [100(6) + 2(200)3]\mathbf{u}_\theta$$
$$= \{-700\mathbf{u}_r + 1800\mathbf{u}_\theta\}\ \text{mm/s}^2$$

The magnitude of $\mathbf{a}$ is

$$a = \sqrt{(700)^2 + (1800)^2} = 1930\ \text{mm/s}^2 \qquad \textit{Ans.}$$

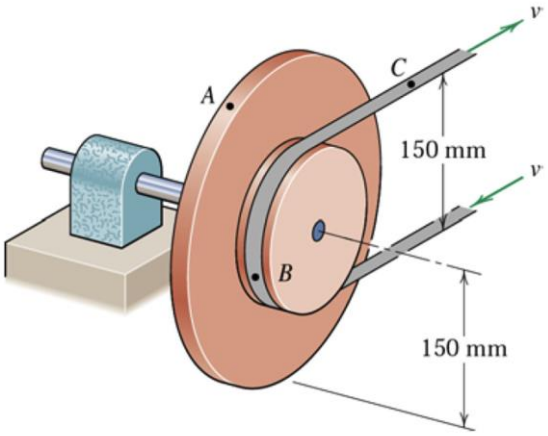
$$\phi = \tan^{-1}\left(\frac{1800}{700}\right) = 68.7° \qquad (180° - \phi) + 57.3° = 169° \qquad \textit{Ans.}$$

NOTE: The velocity is tangent to the path; however, the acceleration is directed within the curvature of the path, as expected.

Question 5

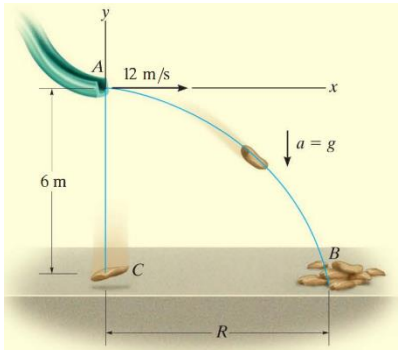
A belt driven pulley rotates with increasing angular velocity. At a certain instant, the belt speed v is $1.5\ m/s$ and the total acceleration of point A is $75\ m/s^2$. Determine the angular acceleration of the pulley, the total acceleration of point B and the acceleration of point C on the belt.

[Ans. $\alpha = 300\ rad/s^2, a_B = 37.5\ m/s^2, a_C = 22.5\ m/s^2$]



Question 6

A sack slides off the ramp, shown in figure, with a horizontal velocity of $12m/s$. If the height of the ramp is $6m$ from the floor, determine the time needed for the sack to strike the floor and the range R where sacks begin to pile up.



Solution

Coordinate System. The origin of coordinates is established at the beginning of the path, point A. The initial velocity of a sack has components $(v_A)_x = 12\ m/s$ and $(v_A)_y = 0$.

Also, between points A and B the acceleration is $a_y = -9.81\ m/s^2$. Since $(v_B)_x = (v)_x = 12\ m/s$, the three unknowns are $(v)_y$, R, and the time of flight t_{AB}. Here we do not need to determine $(v_B)_y$.

Vertical Motion. The vertical distance from A to B is known, and therefore we can obtain a direct solution for t_{AB} by using the equation

$$y_B = y_A + (v_A)_y\ t_{AB} + \left(\frac{1}{2}\right) a_y t_{AB}^2$$

$$-6m = 0 + 0 + \frac{1}{2}\left(-\frac{9.81m}{s^2}\right) t_{AB}^2$$

$$t_{AB} = 1.11s$$

Horizontal Motion. Since t_{AB} has been calculated, R is determined as follows:

$$R = x_B = x_A + (v_A)_x\ t_{AB}$$

$$R = 0 + (12\ m/s)(1.11\ s)$$

$$R = 13.3\ m$$

The calculation for t_{AB} also indicates that if a sack were released from rest at A, it would take the same amount of time to strike the floor at C.